

BUSINESS INSIGHTS REPORT — MARKETING CAMPAIGN EDA

Task 04 | Data Analyst Internship — SkillCraft Technology

Objective

The objective of this analysis was to perform Exploratory Data Analysis (EDA) on marketing campaign data, understand the customer conversion funnel, compare marketing channels, and recommend where additional marketing budget should be allocated based on Return on Investment (ROI).

1. Data Preparation

The dataset contained 505 records and 10 columns. During cleaning, 5 duplicate records were removed. Missing categorical values were handled using the mode, while missing numerical values were filled using the median. Text values were also standardized to ensure consistent analysis.

2. Marketing Funnel Analysis

The marketing funnel was examined across four stages: **Impressions** → **Clicks** → **Leads** → **Conversions**. CTR, lead rate, and conversion rate were calculated to understand movement through the funnel and identify potential drop-off points.

3. ROI Analysis

ROI = ((Revenue – Ad Spend) / Ad Spend) × 100. Email produced the highest ROI at approximately **116.76%**, followed by Google Ads at approximately **106.04%**. Instagram Ads returned about 51.93%, Social Media 40.65%, and YouTube Ads 36.15%. Facebook Ads recorded approximately **–1.15%**, meaning its revenue was slightly below advertising expenditure.

Channel	ROI	ROAS	Budget Action
Email	116.76%	2.17	Increase
Google Ads	106.04%	2.06	Increase
Instagram Ads	51.93%	1.52	Maintain / Optimize
Social Media	40.65%	1.41	Maintain / Optimize
YouTube Ads	36.15%	1.36	Optimize
Facebook Ads	–1.15%	0.99	Review / Reduce

4. Business Recommendations

Based on the EDA and ROI results, **Email and Google Ads should receive a larger share of the marketing budget** because they delivered the strongest returns. Instagram Ads and Social Media should be maintained and optimized. YouTube Ads should be improved before a major budget increase. Facebook Ads should be reviewed and potentially receive a reduced budget until its profitability improves.

5. Conclusion

The analysis shows that effective budget allocation should consider both revenue and marketing efficiency. Prioritizing high-ROI channels while optimizing or reassessing weaker channels can improve overall campaign profitability and make future marketing spending more data-driven.

Tools: Python • Pandas • Matplotlib • Exploratory Data Analysis